

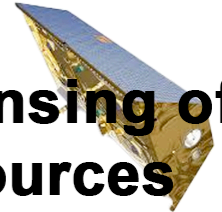


Australian
National
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of Earth Sciences



**Remote Sensing of
Water Resources**
EMSC3025/6025



Aquifers

EMSC3025/6025

Objectives

By the end of this lecture, you should be able to:

Aquifer Classification:

- Define **aquifers** as the fundamental unit of groundwater hydrology
- Distinguish between **unconfined**, **confined**, and **perched** aquifers
- Classify confining beds: **aquifuge**, **aquiclude**, **aquitard**, and real-world **aquitardifers**

Aquifer Properties:

- Calculate **transmissivity** ($T = Kb$) and **storativity** (S)
- Explain **specific storage**, **specific yield**, and **specific retention**
- Compare storage mechanisms in confined vs unconfined aquifers

Objectives

By the end of this lecture, you should be able to:

Aquifer Systems:

- Identify major aquifer types: **basin-fill**, **fluvial**, **semiconsolidated**, **sandstone**, and **carbonate-rock** aquifers
- Understand how depositional environments control aquifer properties
- Recognise **karst features**: sinkholes, solution conduits, and cave systems
- Apply concepts to real-world examples (Dakota Sandstone, Gulf Coast, Floridan aquifer system)

Introduction

- Just as the **catchment** is the basic unit of surface hydrology, the **aquifer** is the fundamental unit of groundwater hydrology.
- Aquifers are defined by their hydraulic properties rather than strictly by geology.
- Understanding aquifer types and properties is crucial for:
 - **Groundwater management**
 - **Well design and sustainability**
 - **Resource assessment**

Key concept

An aquifer's definition depends on context —what's considered an aquifer in one region might be a confining bed in another.

Defining an Aquifer

An aquifer is “a lithologic unit or combination of lithologic units capable of yielding water to pumped wells or springs”

— Domenico (1972)

- An aquifer can be:
 - Coextensive with **geologic formations**
 - A group of formations
 - Part of a formation
 - Independent of geologic units (cutting across formations)

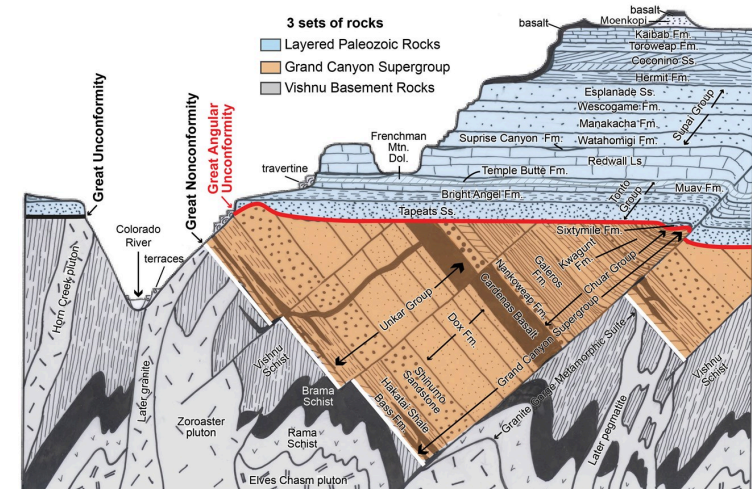


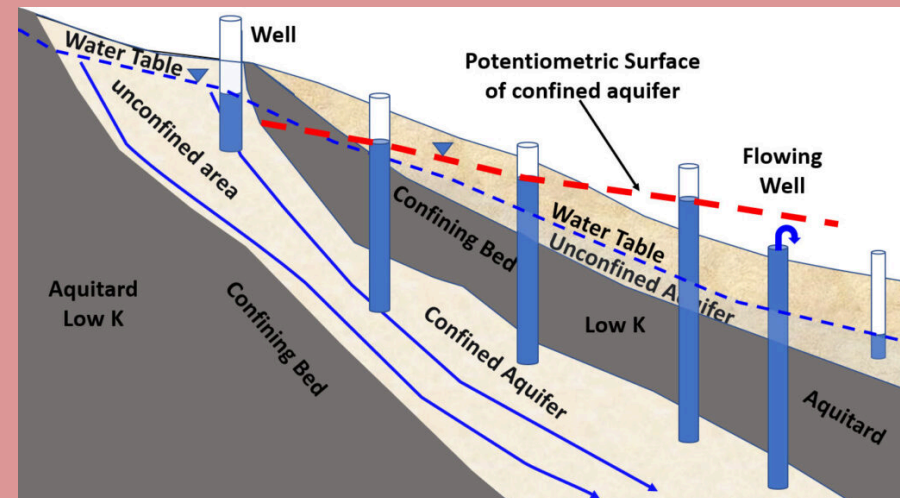
Figure 1: Geologic formations can be very complex, and Aquifers can be coextensive with them.

Confining Beds

- Units of **low permeability** that bound an aquifer are called **confining beds**.
- The designation is **context-dependent**:
 - In areas with prolific aquifers → low permeability unit = confining bed
 - In groundwater-poor regions → same deposit might be considered an aquifer

Important note

In practice, this ambiguity is resolved by explicitly defining **hydraulic conductivity** or **porosity** values that characterise the unit.



Unconfined Aquifers

- Also called **water table aquifers**
- The **water table** forms the upper boundary
- No confining layer above
- Wells or piezometers installed in unconfined aquifers approximately define the water table position
- Water is under **atmospheric pressure** at the water table

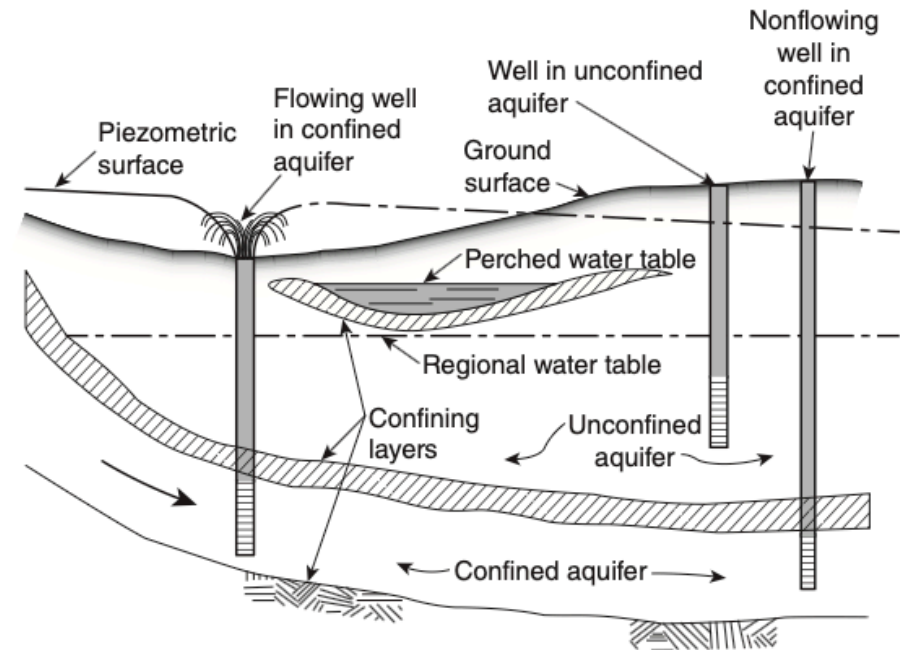


Figure 2: Unconfined, confined, and perched aquifers

Confined Aquifers

- Also called **artesian aquifers**
- Bounded by confining beds on **both top and bottom**
- Water is under **pressure** greater than atmospheric
- The water level in a well occurs **above the upper boundary** of the aquifer
- Occasionally, water level occurs **above ground surface** (flowing artesian well)

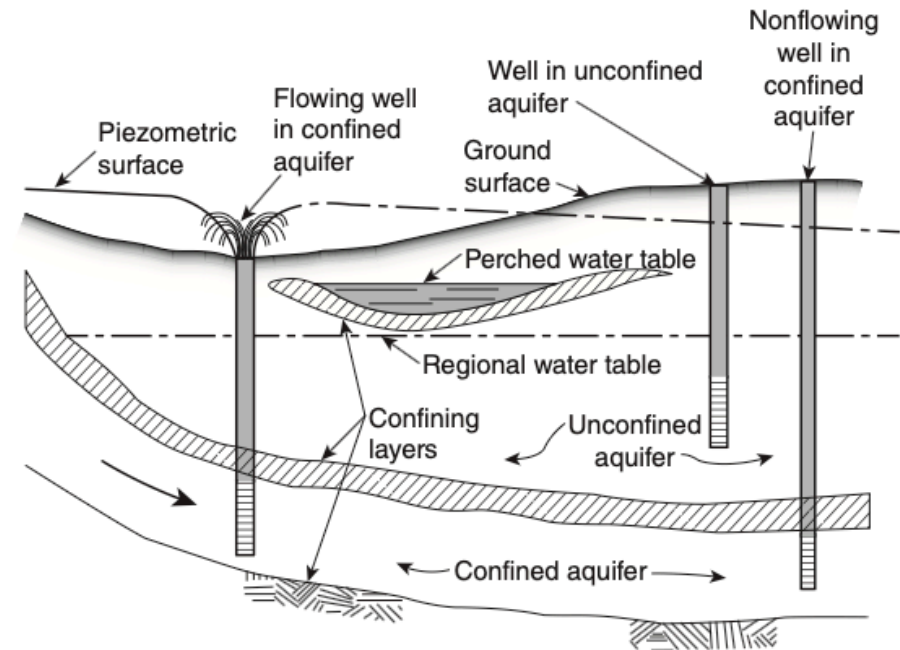


Figure 3: Unconfined, confined, and perched aquifers

Perched Aquifers

- An **unconfined aquifer** that develops **above the regional water table**
- Forms on top of a low hydraulic conductivity layer
- An **unsaturated zone** exists **below** the perched aquifer
- Often temporary or seasonal
- Important for local water supply in some areas

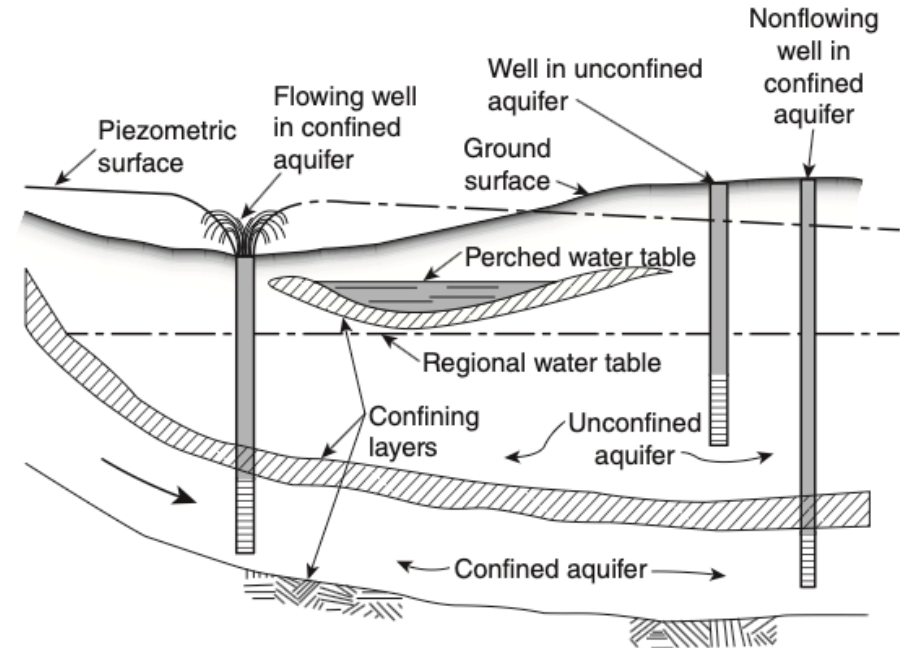


Figure 4: Unconfined, confined, and perched aquifers

Key feature: Unsaturated zone both above and below the perched zone

Comparing Aquifer Types

Aquifer Type	Upper Boundary	Lower Boundary	Pressure	Water Level
Unconfined	Water table (free surface)	Confining bed or impermeable layer	Atmospheric at water table	At water table
Confined	Confining bed	Confining bed	Above atmospheric	Above aquifer top
Perched	Water table	Low-K layer (perching layer)	Atmospheric at water table	Above regional water table

Table 1: Summary of aquifer types and their characteristics

Types of Confining Beds

Different Classifications of Confining Beds

Three terms are occasionally used to describe different types of confining beds:

Aquifuge

- The ultimate low hydraulic conductivity unit
- **Extremely poor** conductor of groundwater
- **No storage capacity**
- Example: intact unfractured crystalline rock

Aquiclude

- Similarly **low permeability** as aquifuge
- **Able to store water** (has porosity)
- Transmits water very slowly
- Example: clay layers

Aquitard

- A **low permeability unit** (leaky confining bed)
- Can **store and transmit** water between adjacent aquifers
- Stored water available to wells in nearby aquifers
- Example: silty clay layers

The Role of Aquitards

Aquitards play important dual roles:

- **Influence rates of recharge** between aquifer layers
- Can **protect underlying aquifers** from contaminants migrating downward from the ground surface
- Act as a **barrier** to vertical contaminant transport
- Provide **slow release** of water to adjacent aquifers during pumping

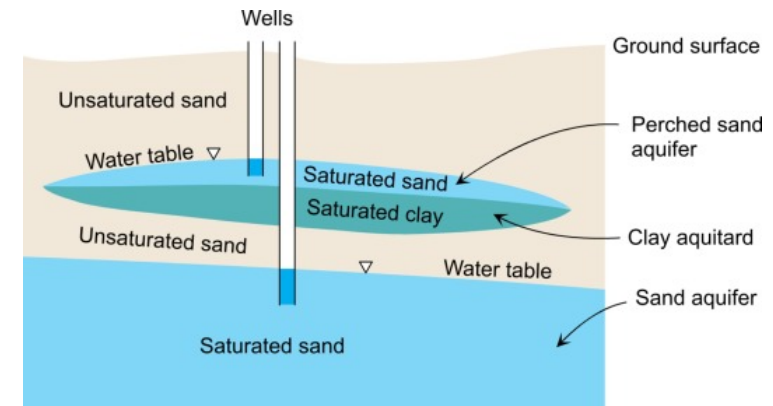


Figure 5: Aquitard between two aquifers

The Complexity of Real Aquitards

- Detailed characterisation in the **midcontinent region of North America** shows that simple concepts don't capture reality.
- Runkel et al. (2018) use the term “**aquitardifers**” to describe confining beds that behave **both as an aquitard and aquifer** at the same place.

Key insight

Real confining beds are often more complex than simple one-dimensional models suggest.

Characteristics of Aquitardifers

Extreme anisotropy in hydraulic conductivity:

- Horizontal direction:
 - Permeable partings **parallel to bedding**
 - Can transmit water horizontally like an aquifer
- Vertical direction:
 - Layering of low permeability units
 - Sufficiently low K_v to inhibit vertical transport of contaminants

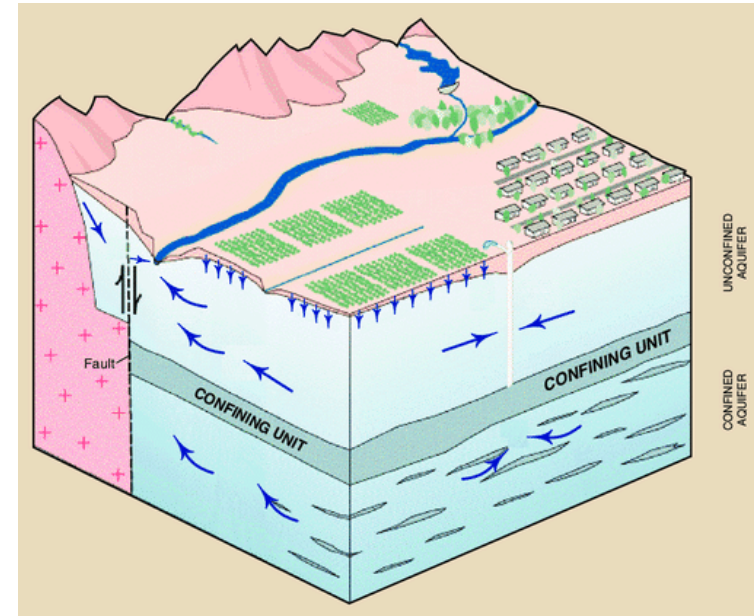


Figure 6: Aquitardifers

Additional Complexity: Fractures

In some shallow settings, aquitardifers develop **vertical fractures**:

- Create **fast-flow pathways**
- Can extend over **relatively large areas**
- Provide direct connection between surface and deeper units
- Completely change the protective function of the confining bed
- Challenge simple predictions of contaminant transport

Management implication

Detailed site characterisation is essential
—simple textbook models may not apply.

Key Aquifer Properties

Two Important Aquifer Properties

When a pump is turned on in a well:

1. Water level in the well casing (and hydraulic head) is **reduced**
2. This causes groundwater to **flow from the aquifer into the well**
3. Much of the pumped water initially comes from “**storage**” in the aquifer

Therefore, aquifers have at least two important characteristics:

- **Ability to store** groundwater
- **Ability to transmit** this water to a nearby well

Key concept

These properties depend to an important extent on the **geologic setting**.

Transmissivity

Definition

The rate at which water of prevailing kinematic viscosity is transmitted through a **unit width** of the aquifer under a **unit hydraulic gradient**.

- Similar concept to hydraulic conductivity
- Main difference: applies across the **vertical thickness** of an aquifer
- Integrates conductivity over aquifer thickness

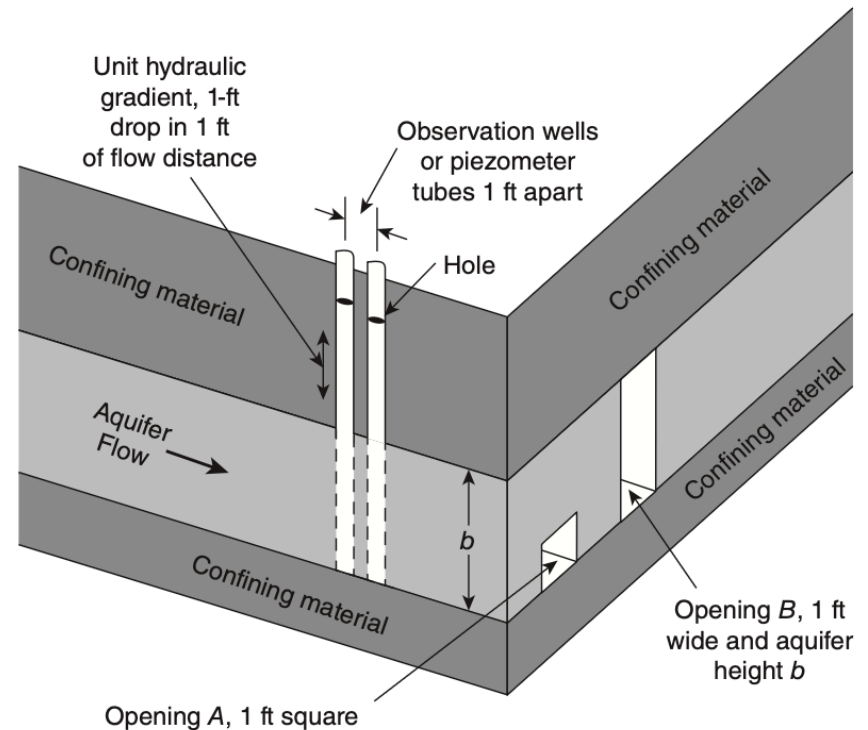


Figure 7: Transmissivity and Conductivity

Transmissivity Equation

If the thickness of the aquifer is b , the transmissivity T is:

$$T = K \times b$$

Where:

- T = transmissivity [L^2/T]
- K = hydraulic conductivity [L/T]
- b = aquifer thickness [L]

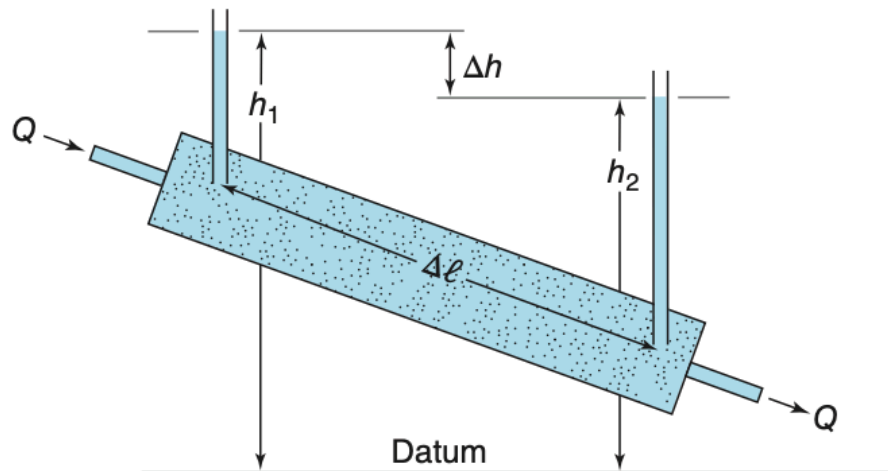
Units: m²/day, ft²/day, or gpd/ft (gallons per day per foot)

Physical meaning

Transmissivity tells you how much water can flow through the **entire thickness** of an aquifer per unit width.

Darcy's Law for Aquifers

For a **homogeneous confined aquifer**, Darcy's equation can be written in terms of transmissivity:



Apply Darcy's law across the aquifer:

$$Q = T \times W \times i$$

Where:

- Q = discharge [L^3/T]
- T = transmissivity [L^2/T]
- W = width perpendicular to flow [L]
- i = hydraulic gradient [dimensionless]

Storativity and Specific Storage

Aquifers can store water. **How this storage is accomplished differs** depending upon whether the aquifer is confined or unconfined.

Confined aquifer — Two mechanisms supply water:

1. **Water expansion:** Lower pressure $\rightarrow$ water molecules expand slightly
2. **Aquifer compression:** Lower water pressure $\rightarrow$ less support for skeleton $\rightarrow$ grains pushed closer together $\rightarrow$ squeezes water out

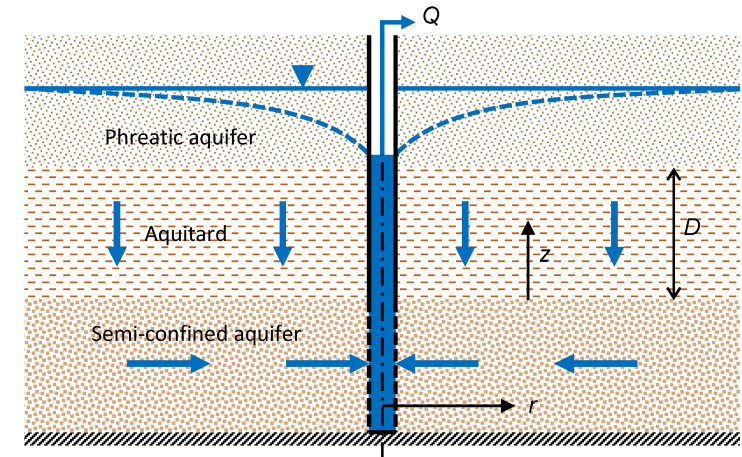


Figure 8: Confined aquifer before and after pumping

Understanding Aquifer Compression

Before pumping:

- Water pressure holds grains apart
- Aquifer skeleton is supported
- Pore space is at maximum

During pumping (head declines):

- Water pressure decreases
- Weight of overlying rocks pushes down
- Grains move closer together
- Pore space reduces → water squeezed out

Result: Only **small amounts** of water released ($S = 10^{-5}$ to 10^{-3})

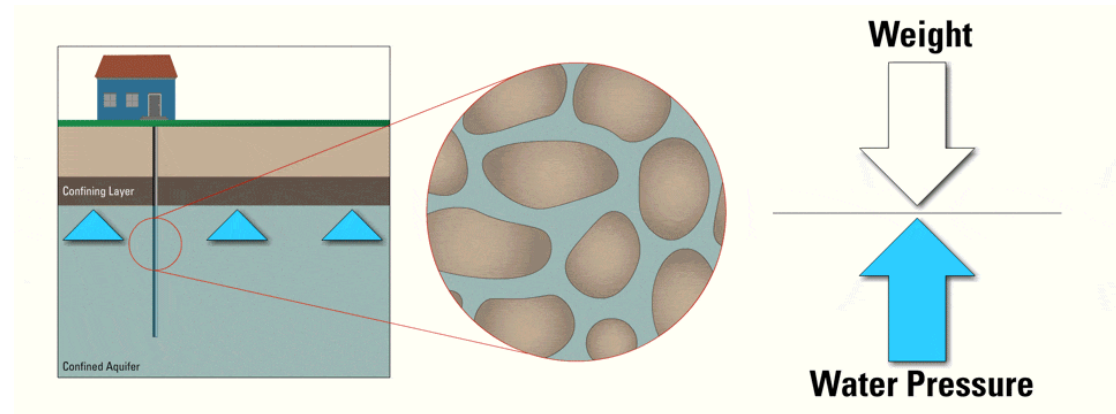


Figure 9: Aquifer compaction

Land Subsidence: Permanent Compression

When pumping is **excessive or prolonged**:

- Initial compression is **elastic** (recoverable)
- Continued compression becomes **plastic** (permanent)
- Aquifer structure permanently altered
- Land surface subsides

Why clay layers are critical:

- Clay is ~1000× more compressible than rock
- Major contributor to subsidence
- Once compressed, **cannot recover**

Real-world example

Houston, Texas:

- Groundwater depletion from Gulf Coast aquifer
- Land subsided **>9 ft (>2.75 m)**
- Damage to infrastructure
- Increased flooding risk

Storage in Unconfined Aquifers

Unconfined aquifer:

- Main source of water is **drainage of water from pores** as the water table declines
- Much more water released per unit decline in head
- For a comparable unit decline in hydraulic head, an unconfined aquifer releases **much more water** from storage than a confined aquifer

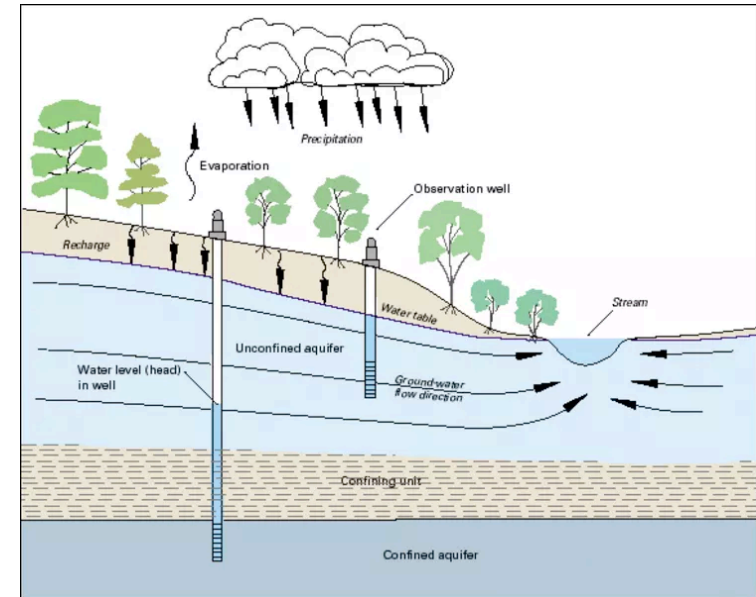


Figure 10: Unconfined aquifer with water table dropping, water draining from pores above the new water table level

Storativity Definition

Storativity (S) is the volume of water that an aquifer releases from or takes into storage per **unit surface area** of the aquifer per **unit change in head**.

$$S = \frac{\text{volume of water}}{\text{unit area} \times \text{unit head change}} = \frac{m^3}{m^2 \times m}$$

- Dimensionless quantity
- For **confined aquifers**: S ranges between 10^{-3} to 10^{-5}
- For **unconfined aquifers**: S (or specific yield) ranges between 0.01 to 0.30

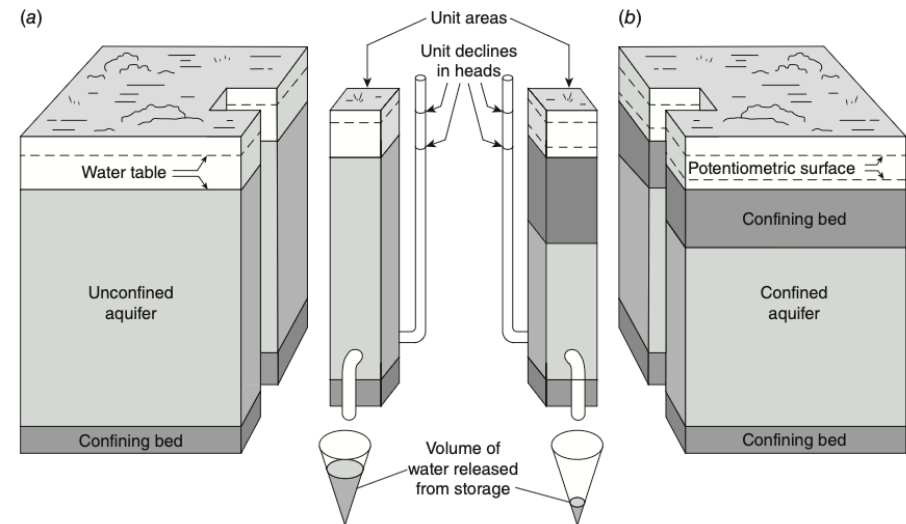


Figure 11: Concept of storativity and specific storage

Specific Storage

Specific storage (S_s) is the volume of water that an aquifer releases from or takes into storage per **unit volume** of aquifer per **unit change in head**.

$$S_s = \frac{\text{volume of water}}{\text{unit volume} \times \text{unit head change}}$$

Units: $[1/L]$ (e.g., $1/m$ or $1/ft$)

Relationship between S and S_s :

$$S = S_s \times b$$

where b is the aquifer thickness.

Visual Summary: Storativity

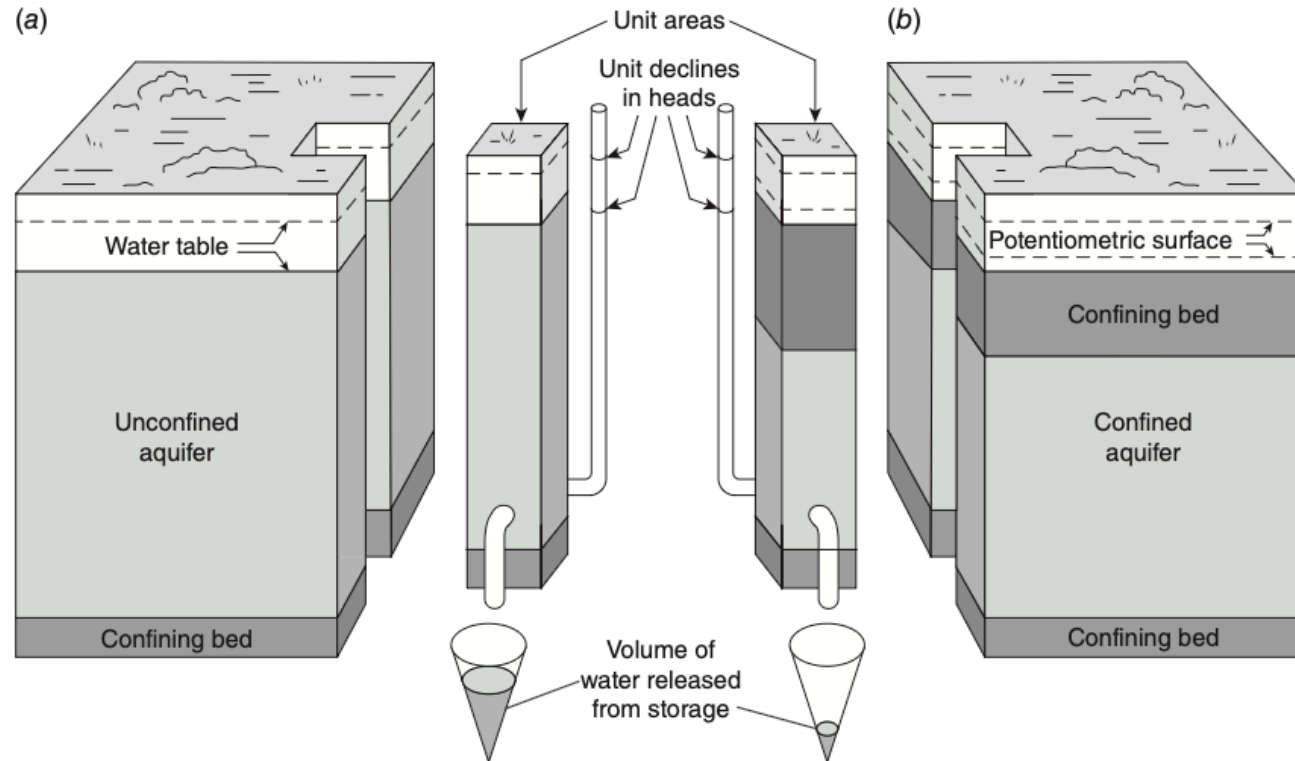


Figure 12: Concept of storativity: confined aquifer shows small release due to water expansion and aquifer compression; unconfined aquifer shows large release due to drainage from pores

Comparing Storage Properties

Property	Symbol	Definition	Units	Typical Values
Specific Storage	S_s	Storage per unit volume	$[1/L]$	10^{-6} to 10^{-4} (1/m)
Storativity (confined)	S	Storage per unit area	Dimensionless	10^{-5} to 10^{-3}
Specific Yield (unconfined)	S_y	Drainable porosity	Dimensionless	0.01 to 0.30

Table 2: Summary of aquifer storage properties

Mathematical Formulation of Storage

Specific Storage in Confined Aquifers

For a confined aquifer, specific storage reflects storage from:

- **Compression of the granular matrix**
- **Expansion of water**

The mathematical definition (Domenico and Schwartz, 1998):

$$S_s = \rho_w g (\beta_p + n \beta_w)$$

Where:

- ρ_w = density of water [ML^{-3}]
- g = gravitational constant (9.81 m/sec²) [LT^{-2}]
- n = porosity of the aquifer
- β_p = vertical compressibility of rock matrix
- β_w = compressibility of water

Compressibility of Water and Matrix

Compressibility (β) quantifies how much a material compresses under pressure.

Units: inverse of pressure [$1/Pa$] or [m^2/N]

Key insight:

- **Higher** compressibility = more compressible
- **Lower** compressibility = more rigid/incompressible

Water compressibility (β_w):

- $4.4 \times 10^{-10} \text{ m}^2/\text{N}$ at 25°C

Material property comparison

Clay is ~1000× more compressible than sound rock

This is why clay layers are major contributors to land subsidence!

Compressibility Values for Earth Materials

Geologic Materials (matrix compressibility α)

Material	Compressibility (m^2/N)
Clay	1×10^{-6} to 1×10^{-8}
Sand	1×10^{-7} to 1×10^{-9}
Gravel	1×10^{-8} to 1×10^{-10}
Jointed Rock	1×10^{-8} to 1×10^{-10}
Sound Rock	1×10^{-9} to 1×10^{-11}

Table 3

Fluid compressibility β

Fluid	Compressibility (m^2/N)
Water	4.4×10^{-10}

Table 4

Note: Loose sediments are more compressible than well-cemented rocks

Using the Specific Storage Equation

The equation $S_s = \rho_w g(\beta_p + n\beta_w)$ can be used to estimate specific storage and storativity.

Example calculation:

- Clay layer with $n = 0.3$
- $\beta_p = 1 \times 10^{-7} \text{ m}^2/\text{N}$
- $\beta_w = 4.4 \times 10^{-10} \text{ m}^2/\text{N}$
- $\rho_w g \approx 10^4 \text{ N/m}^3$

$$S_s = 10^4(1 \times 10^{-7} + 0.3 \times 4.4 \times 10^{-10})$$

$$S_s \approx 1 \times 10^{-3} \text{ m}^{-1}$$

Key observation

The matrix compressibility (β_p) term typically **dominates** over the water compressibility term, especially in fine-grained materials like clay.

Storage in Unconfined Aquifers

In an unconfined aquifer, the groundwater response to pumping is **different** from a confined aquifer:

Early time (no significant water level change):

- Water comes from **expansion** of the water
- Water comes from **compression** of the grains
- Similar to confined aquifer behaviors

Later on (water table falls):

- Water mainly comes from **gravity drainage** of pores
- This is the dominant storage mechanism

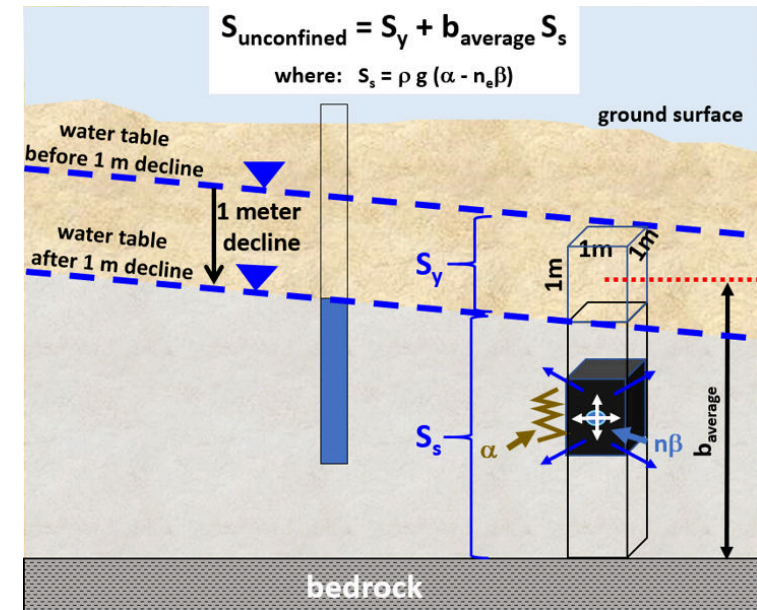


Figure 13: Response to reduction in hydraulic head

Storativity of Unconfined Aquifers

The storativity of an unconfined aquifer is expressed as:

$$S = S_y + bS_s$$

Where:

- S_y = specific yield of the aquifer
- b = aquifer thickness
- S_s = specific storage

Key insight

- Specific yield: 0.1 to 0.3
- Product bS_s : 10^{-3} to 10^{-5}

Thus, **specific yield is the dominant storage term** for unconfined aquifers.

Aquifer Type Transitions

In some cases, an aquifer may be:

- **Confined at early stage** of pumping
- **Becomes unconfined** at late time

Process:

1. Initial pumping creates cone of depression
2. Water levels decline
3. Head falls below the top of the aquifer
4. Aquifer dewateres from the top down
5. Storage mechanism changes from elastic to gravity drainage

As the aquifer changes from confined to unconfined, **storativity values change accordingly.**

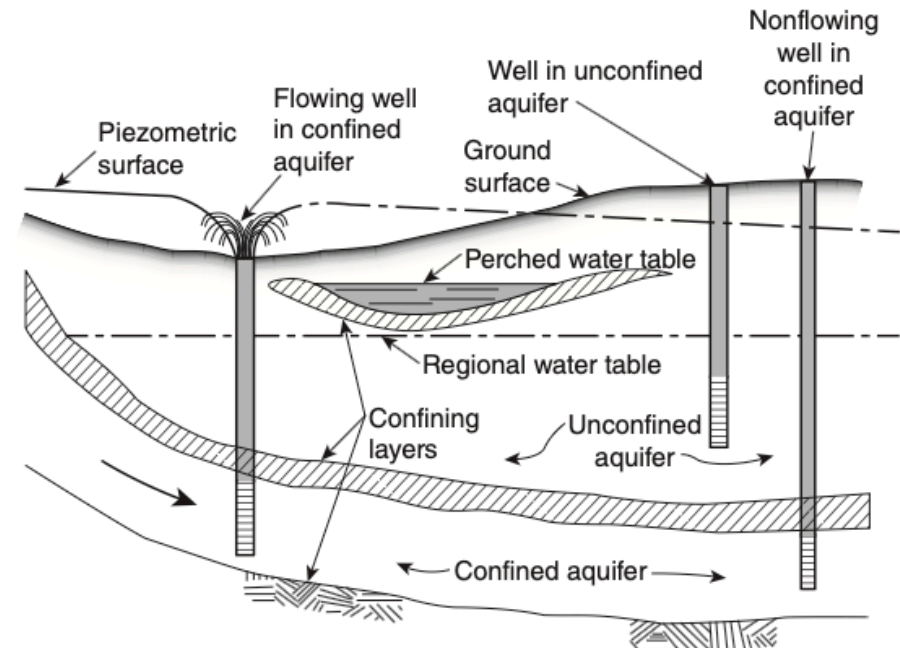


Figure 14: Aquifers can transition from confined to unconfined

Specific Yield and Specific Retention

Specific yield (S_y) is the water released from a water-bearing material by **gravity drainage**.

Say the ratio of the volume of water yielded from a soil or rock by gravity drainage after being saturated, to the total volume of the soil or rock.

Expressed as the ratio of:

$$S_y = \frac{V_d}{V_T}$$

Where:

- V_d = volume of water that drains by gravity
- V_T = total volume of soil or rock (after being saturated)

Specific Retention

Not all water initially present is released from storage.

Specific retention (S_r) describes water that is **retained** as a film on grain surfaces or held in small openings by molecular attraction.

$$S_r = \frac{V_r}{V_T}$$

Where:

- V_r = volume of water retained against gravity
- V_T = total volume of soil or rock

Important relationship

Porosity is related to specific yield and specific retention by:

$$n = S_y + S_r$$

The sum equals the total porosity.

Specific Retention Characteristics

Specific retention **increases with:**

- **Decrease in grain size** (more surface area)
- **Decrease in pore size** (stronger capillary forces)

Typical values:

- Coarse gravel: $S_r \approx 0.01$ to 0.05
- Fine sand: $S_r \approx 0.05$ to 0.15
- Silt: $S_r \approx 0.15$ to 0.30
- Clay: $S_r \approx 0.30$ to 0.50

Inverse relationship

Materials with high specific retention have **low specific yield**, and vice versa.

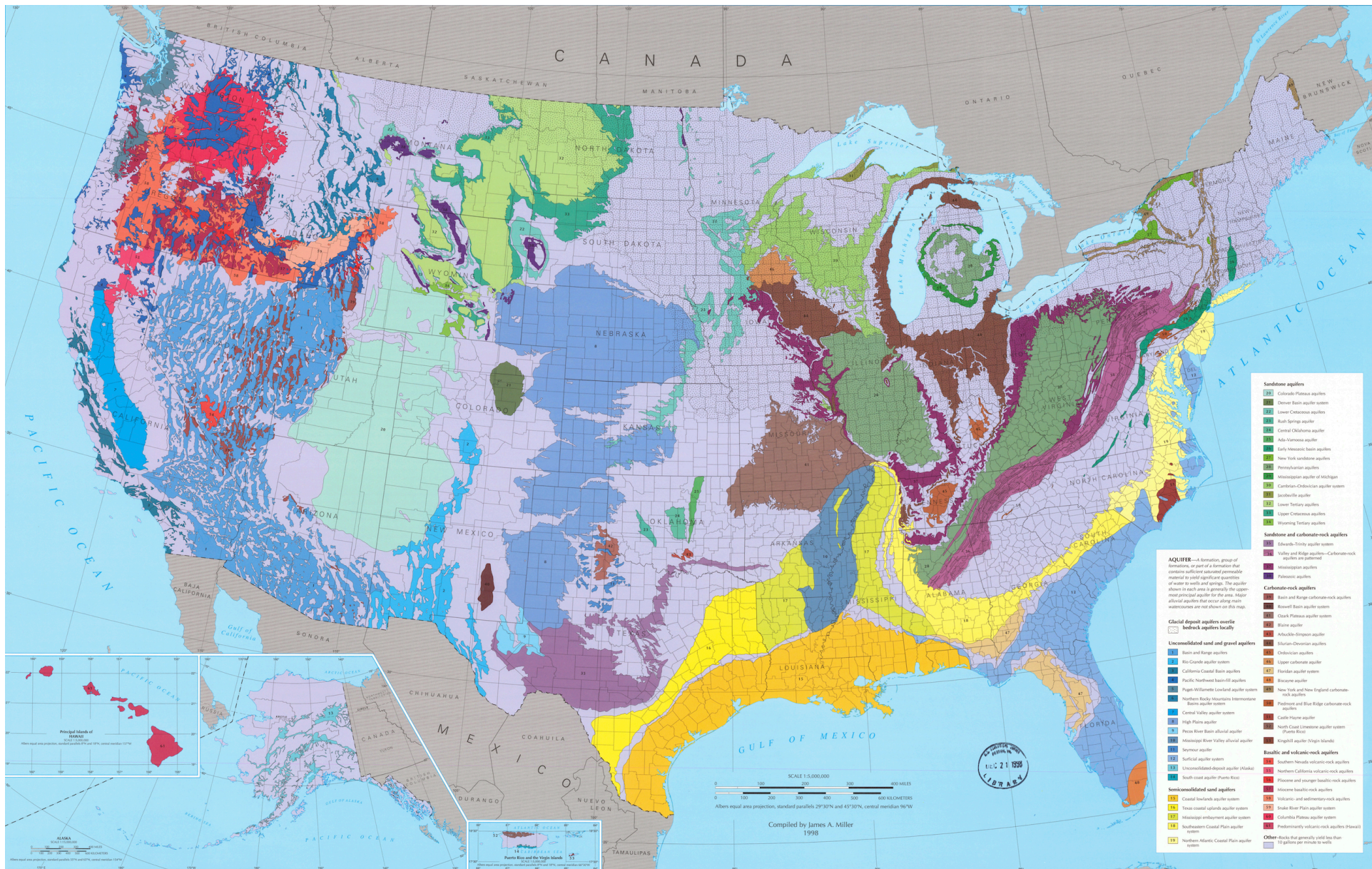
This is why clay has high porosity but yields little water!

Examples of Real-World Aquifer Systems

[Back to water-course](#)

EMSC3025/6025: Remote Sensing of Water
Resources

Dr. Sia Ghelichkhan



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Groundwater in Australia

Page last updated: 27 June 2014

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Groundwater is found in aquifers which are geological formations able to store and transmit water. The geology of Australia is diverse, therefore the types of aquifers vary, examples include:

[Alluvial Aquifers](#) - Formed of sediments such as gravel, sand and silt deposited by rivers or other running water.

Groundwater in Australia

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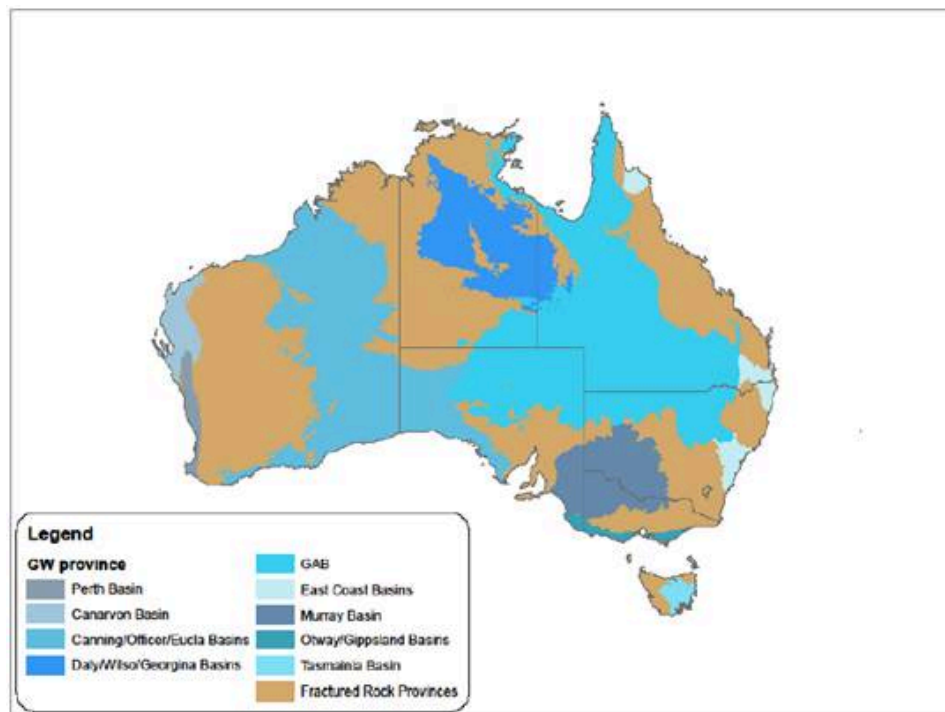
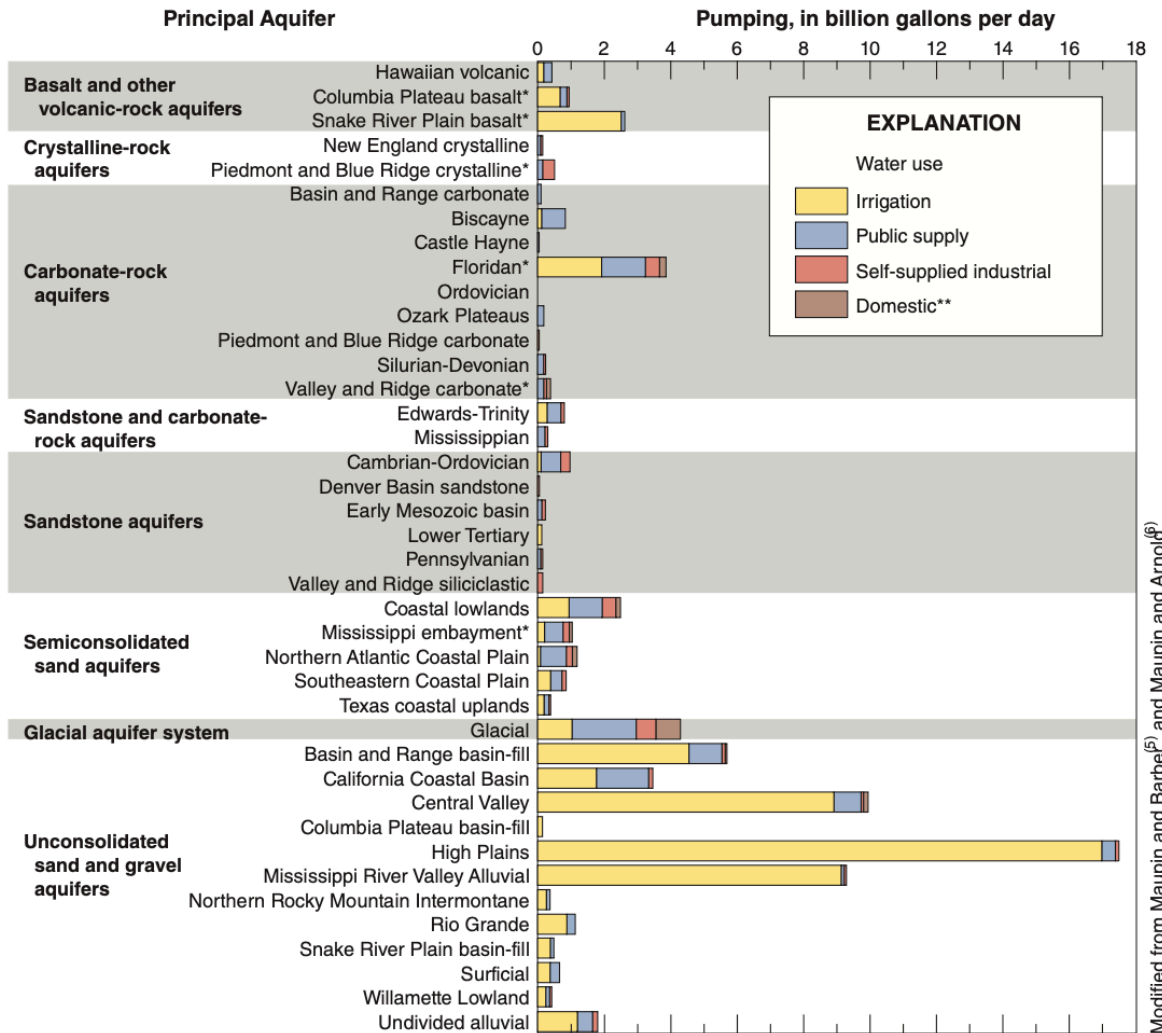


Figure 16: Principal aquifers in the United States

Principal Aquifers in the United States



*Includes domestic use for an adjacent aquifer in this assessment.

**Domestic water use was not available for all aquifers.

Principal Aquifers in the United States organized by aquifer type and showing pumping rates for different uses

- This collection accounts for about 90% of daily groundwater production in the US.

Unconsolidated Sand and Gravel Aquifers

Characteristics:

- Intergranular porosity
- Primarily **unconfined** conditions
- High permeability (varies with clay content and sorting)
- Most extensively pumped aquifer type in US

Four main categories:

1. **Basin/valley-fill:** Central Valley (California), Arizona basins
2. **Blanket sand and gravel:** High Plains aquifer (most pumped)
3. **Glacial deposits:** North of glaciation line (outwash, terraces)
4. **Stream-valley aquifers:** Local, small extent (not mapped)

Key properties

- High storage and transmission
- Thicker deposits → more regional flow
- **Susceptible to contamination**
- Provide much of western US groundwater

US Distribution: Sand and Gravel Aquifers

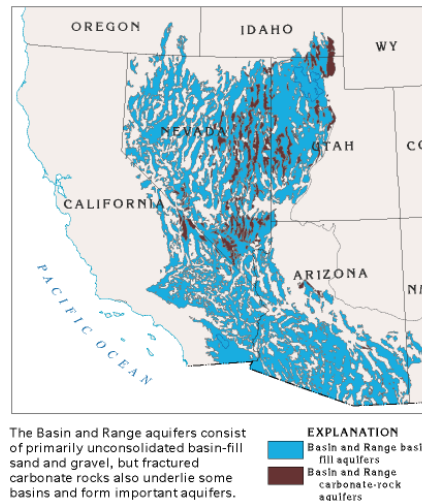
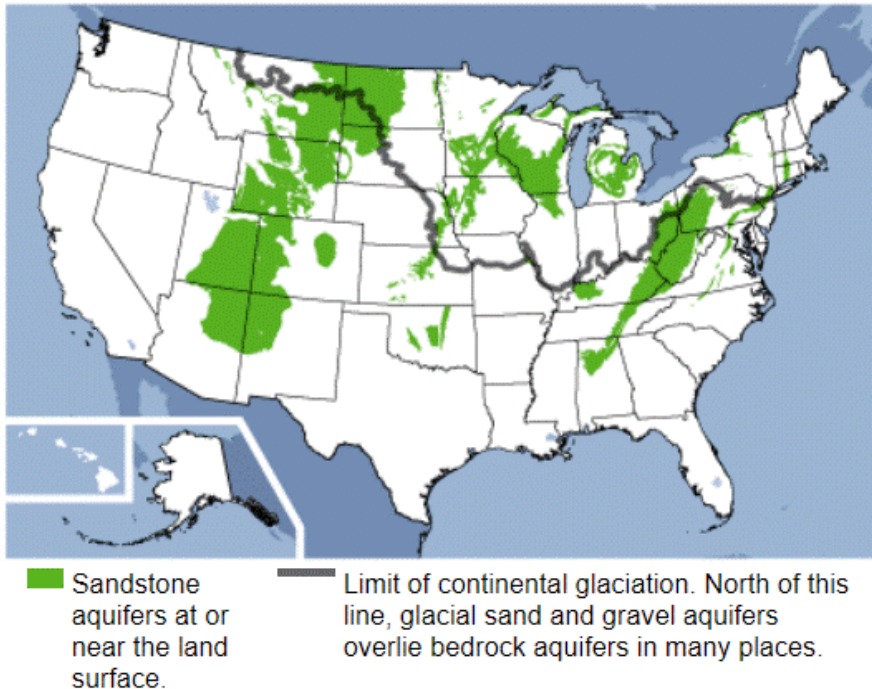


Figure 17: Basin-fill aquifers in Basin and Range Province

Basin-fill aquifers concentrated in Basin and Range Province (western US)

US Distribution: Sandstone Aquifers

PRINCIPAL SANDSTONE AQUIFERS



Key characteristics:

- Consolidated sand → **reduced porosity**
- Joints and fractures transmit most groundwater
- **Low to moderate** hydraulic conductivity
- Extend over **large areas** → provide large water quantities

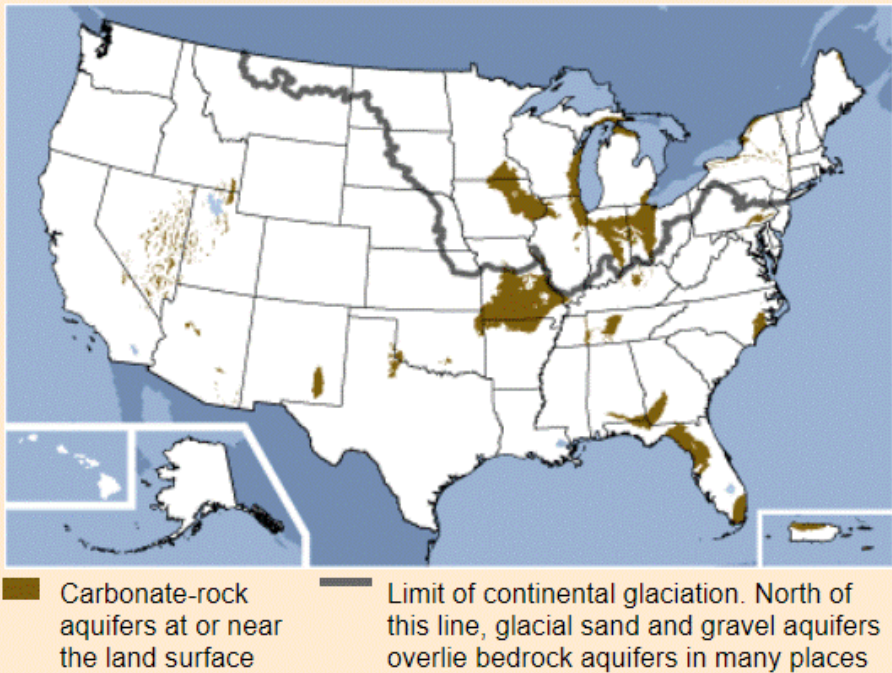
Map note

Shows only shallowest principal aquifer. Deeper, sometimes more productive aquifers may exist below.

Figure 18: Principal sandstone aquifers in the United States

US Distribution: Carbonate-Rock Aquifers

PRINCIPAL CARBONATE-ROCK AQUIFERS



Composition:

- **Calcite** (CaCO_3) → limestone
- **Dolomite** ($\text{CaMg}(\text{CO}_3)_2$) → dolostone
- Marine origin (shells, corals, algae, precipitates)

Key processes:

- Post-depositional: compaction, cementation, dolomitisation
- **Solution enhancement:** dissolution by acidic groundwater
- Creates tubes, widened joints, and caverns

Figure 19: Principal carbonate-rock aquifers in the United States

Carbonate Aquifer Characteristics

Solution openings range from:

- Small tubes and widened joints
- To caverns **tens of meters wide**
- Extending **hundreds to thousands of meters** in length

Hydraulic properties:

- Well-connected networks → **large yields** to wells
- Undissolved rock between openings → nearly impermeable
- Extreme heterogeneity

Famous example

Floridan aquifer system: One of the most productive aquifer systems in the world, supplies water to millions in Florida

Other Important Aquifer Types

Crystalline-rock aquifers:

- Fractured granite, basalt, metamorphic rocks
- Very low primary porosity
- Flow occurs **primarily in fractures**
- Secondary porosity controls productivity
- Important in New England, Canadian Shield
- Well yields highly variable

Volcanic-rock aquifers:

- Basalt flows with **high permeability**
- Flow in cooling fractures and lava tubes
- Vesicular zones store water
- Examples: Hawaiian volcanic, Columbia Plateau
- Can be extremely productive locally
- Complex interbedded structure

Summary of Aquifer Systems

Unconsolidated Aquifers Summary

Basin-Fill Aquifers:

- Structural basins in mountain regions
- Hundreds to thousands of meters thick
- Examples: Basin and Range, Central Valley
- Long, narrow bodies with interbedded materials
- Major water sources in western US

Fluvial Aquifers:

- Deposited by river systems
- Two types: meandering vs braided
- Braided → more continuous and productive
- Long, narrow, thin geometry
- Mississippi River Valley, High Plains

Key characteristics:

- High permeability (varies with sorting)
- Unconfined conditions
- High specific yield (0.1 – 0.3)
- Susceptible to contamination
- Most extensively pumped in US

Categories:

- Basin/valley-fill
- Blanket sand and gravel
- Glacial deposits
- Stream-valley aquifers

Consolidated Aquifers Summary

Semiconsolidated Sediments:

- Gulf Coast aquifer systems
- Wedge-shaped geometry (thicken toward coast)
- Confined updip, unconfined downdip
- Risk: saltwater intrusion, land subsidence
- Example: Houston subsided >9 ft (>2.75 m)

Sandstone Aquifers:

- Cemented sand with reduced porosity
- Flow in fractures and bedding planes
- Low to moderate hydraulic conductivity
- Large areal extent
- Example: Dakota Sandstone (~46% recharge through confining beds)

Carbonate-Rock Aquifers:

- Limestone and dolostone
- Solution-enhanced permeability
- **10 orders of magnitude** range in K
- Large yields where well-connected
- Example: Floridan aquifer system

Other Types:

- **Crystalline-rock:** fractured granite/basalt, flow in fractures
- **Volcanic-rock:** basalt with cooling fractures and lava tubes

Karst Systems Summary

Formation requirements:

1. Chemically aggressive groundwater
2. Fractures present for transmission
3. Groundwater can drain out

Solution features:

- Start as small conduits (~1 cm)
- Grow to cave size (kilometers long)
- Highest yields at fracture intersections
- Extremely heterogeneous permeability

Karst landforms:

- Sinkholes (dissolution, cover subsidence, cover-collapse)
- Stream sinks (disappearing streams)
- Dry valleys
- Large spring systems
- Underground cave networks

Management challenges:

- Rapid contaminant transport
- Unpredictable flow paths
- Surface-subsurface connectivity

Comparing Major Aquifer Types

Aquifer Type	Porosity	K Range	Storage	Key Feature	Main Challenge
Unconsolidated	High (0.25-0.35)	High	$S_y = 0.1 - 0.3$	Intergranular	Contamination
Basin-fill	High	High-Moderate	High	Thick accumulations	Over-exploitation
Fluvial (braided)	High	High	High	Lateral continuity	Variable quality
Semiconsolidated	Moderate	Moderate	Moderate	Wedge geometry	Subsidence, saltwater
Sandstone	Low-Moderate	Low-Moderate	$S = 10^{-3} - 10^{-5}$	Fracture flow	Vertical leakage
Carbonate	Variable	10 orders of magnitude	Variable	Solution conduits	Heterogeneity
Karst	Highly variable	Extremely high locally	Low-Moderate	Caves and conduits	Rapid transport
Crystalline	Very low	Very low	Very low	Fractures only	Limited yield

Table 5

Summary

Key takeaways from this lecture:

Aquifer Classification:

- Aquifers are the fundamental unit of groundwater hydrology
- Three types: **unconfined**, **confined**, **perched**—each with distinct boundary conditions
- Confining beds range from impermeable (aquifuge) to leaky (aquitard), with complex real-world **aquitardifers**

Aquifer Properties:

- **Transmissivity** ($T = Kb$) quantifies ease of transmission; **storativity** (S) quantifies storage release
- Storage mechanisms: elastic (S_s) in confined, gravity drainage (S_y) in unconfined aquifers
- Relationship: $n = S_y + S_r$ connects porosity to drainable and retained water

Summary

Key takeaways from this lecture:

Real-World Aquifer Systems:

- **Basin-fill** (Basin and Range): structural basins with thick sediment accumulation
- **Fluvial**: braided streams create more continuous aquifers than meandering systems
- **Semiconsolidated** (Gulf Coast): wedge-shaped geometry, saltwater intrusion risk, land subsidence issues
- **Sandstone** (Dakota): vertical leakage through confining beds dominates recharge (~46%)
- **Carbonate-rock**: solution enhancement creates 10 orders of magnitude range in hydraulic conductivity
- **Karst landscapes**: dissolution features create unique surface and subsurface drainage systems

Understanding aquifer diversity enables effective groundwater management and